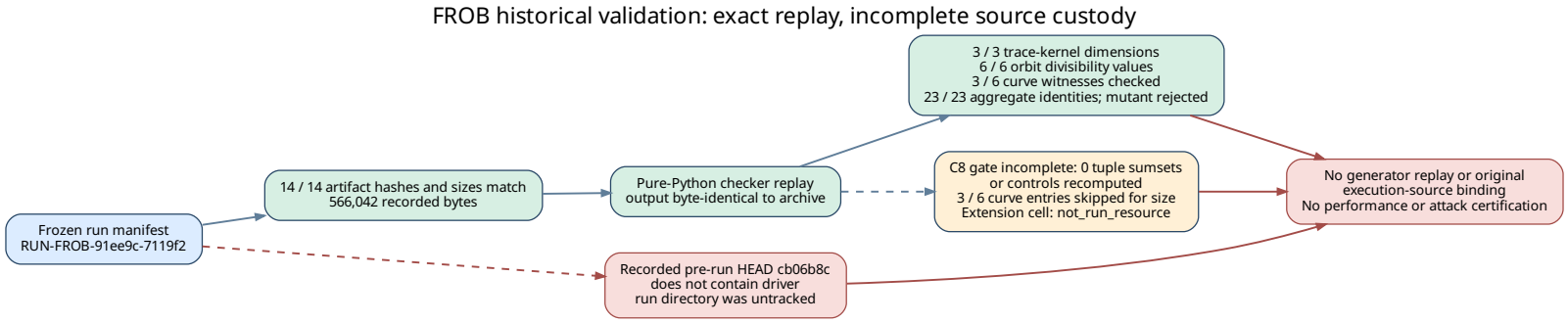
Historical FROB checker replay and source custody

Validation audit, 22 October 2026

Scope. This checks frozen run `RUN-FROB-91ee9c-7119f2` at source tree `e83bb2f14414c9669360816aeee5876dfcfd4384`. It is an integrity audit, not a new FROB measurement or a research verdict. The run directory was copied before executing its pure-Python checker; frozen bytes were not edited.

Check	Result	Limit
Archived artifact hashes and sizes	14/14 match; 566,042 bytes	Storage integrity only
Checker output	Byte-identical replay	Same archived checker source
Trace-kernel dimension; orbit divisibility	3/3; 6/6	Recorded core cells
Curve point/order/Frobenius witnesses	3/6 checked	Other three skipped for size
Aggregate rational identities	23/23; mutant rejected	Counts treated as inputs
Tuple sumsets and C2/C3/C4 arms	0 recomputed	Checker source confirms this limit
Extension cell	`not_run_resource`	540-second historical checkpoint



Source custody. The manifest records pre-run HEAD `cb06b8c20c2d6559041c45c4a4b29dd7f132fdf5` and says the run directory was untracked at execution start. The driver is absent from that commit. Current-schema `code.commit`, `code.command`, and `environment` fields are absent, although separate command and environment artifacts are hash-pinned. Their presence does not bind the executed driver to the recorded HEAD.

Boundary and next check. The archived checker's `tuple_sum_reverifications` label overstates its code: its three `ok` entries check curve witnesses, not tuple sums or control arms. The frozen C8 control requires every tuple sum, membership, order and rational identity to be checked independently, so C8 and the experiment success gate remain unestablished. The replay does not rerun the Sage generator or any performance experiment. Recover a contemporaneous execution snapshot and exact command if available; otherwise run a fresh source-bound replay against the frozen specification and controls. The Coordinator must adjudicate the original record without assigning the pre-run HEAD as a false execution-source commit. Machine receipt: `docs/reports/2026-10-22-frob-provenance-audit/audit.json`.